

Validation (working document)

Liza Buriak^{1*}

^{1*}Institute for Ecological Economics, Vienna University of Economics
and Business (WU), Vienna, Austria.

Corresponding author(s). E-mail(s): TODO@wu.ac.at;

Abstract

Standalone draft of the Validation section. The text is self-contained; appendix pointers refer to the Methods working document (methods.v2).

1 Validation

1.1 Agreement with the Trase benchmark

The reconstructed origin-to-importer flows are evaluated against the independent Trase Brazil-soy dataset (v2.6.1), which links municipalities of production to countries of first import using per-shipment customs and logistics records [1, 2]. All three allocation variants are compared against this benchmark on the municipality×destination flow matrix: the downscaling baseline, the Euclidean model and the multimodal transport variant. Pooled over all flows the three perform comparably, with mean correlations of 0.71, 0.69 and 0.69 over 2010–2022. The value added by the supply-chain structure shows at the destination level: the mean per-destination correlation is 0.41 for the Euclidean model and 0.42 for the multimodal variant, against 0.28 for downscaling. Sample construction and metrics are given in the validation appendix of the Methods document.

1.2 Agreement grows with flow size

Pooled over the 127,862 matched municipality×destination×year flows, the log-flow correlation with Trase is 0.68 and 54% of the flows fall within a factor of three of the benchmark value. In the upper half of the volume distribution (flows above 1.2 kt, which carry 98% of the benchmark tonnage) the within-factor-three share rises to 56% and the median deviation falls from a factor of 2.7 to 2.4; per destination, the mean correlation rises from 0.29 in the lower half of import volumes to 0.53 in the upper half. The tonnage that carries the footprint is therefore also the best-validated part of the flow distribution, and below roughly 10^3 t per destination subnational modelling adds no information over downscaling. This motivates the volume floor of the reporting rule: model subnationally above the floor, downscale below it, and read individual small-volume destinations with caution.

1.3 Backend consistency over time

Before 2010 the pipeline runs on FABIO v1.1 inputs instead of the v2 build. In the clean overlap years the two backends differ by 6–8% in the total footprint with cross-country correlations above 0.988, so pre-2010 levels are reported unadjusted.

1.4 Sensitivity to input assumptions

The two least constrained input assumptions are screened for their effect on the benchmark agreement: the within-state location of crushing (capacity is reported by state and the baseline divides it equally among each state’s plants) and the feed and livestock parametrisation (per-system feed rates and the 2010 herd composition). Twenty-four perturbed variants of the flow reconstruction are re-run through the full chain up to the origin-to-importer flows for a representative year (2019) and scored against Trase on the pooled municipality×destination correlation (Euclidean model, base $r = 0.72$). For the crush assumption, each state’s fixed capacity total is redistributed across its plant municipalities in proportion to municipal soybean production or by ten seeded random draws over unknown plant sizes, and the allocation weight is sharpened or flattened (capacity raised to $\gamma \in \{0.5, 2\}$). For the feed assumption, the per-system bean and cake feed rates receive a lognormal jitter of $\sigma = 0.3$ across ten seeded draws (a uniform multiplier cancels in the national rescale, so the jitter shifts only the relative system mix), and the 2010-frozen municipal herd composition is replaced by state-average shares.

The two assumptions separate cleanly (Table 1, Figure 1). Crush location accounts for nearly all of the induced variation: the random within-state splits span correlations of 0.67 to 0.74 (spread 0.073), the production-proportional split scores 0.69, and the allocation-sharpness variants span 0.024. The feed variants are immaterial: the ten feed-rate draws move the correlation by less than 0.01 and the herd-composition variant by less than 0.001. Agreement with the benchmark is therefore limited by where processing sits within states, not by the feed model, and even the widest perturbation leaves the correlation within 0.05 of the reported value.

Table 1 Input-assumption sensitivity of the model–Trase correlation (pooled municipality×destination flows, Euclidean model, 2019; base $r = 0.723$).

Assumption	Perturbation	Runs	r range	Spread
Crush location	random within-state splits	11	0.671–0.744	0.073
Crush location	allocation sharpness $\gamma \in \{0.5, 2\}$	2	0.706–0.731	0.024
Feed rates	per-system lognormal jitter ($\sigma = 0.3$)	10	0.717–0.725	0.007
Herd composition	state-average system shares	1	0.722	< 0.001

References

- [1] Godar, J., Persson, U. M., Tizado, E. J. & Meyfroidt, P. Towards more accurate and policy relevant footprint analyses: Tracing fine-scale socio-environmental impacts of production to consumption. *Ecological Economics* **112**, 25–35 (2015).
- [2] Escobar, N. *et al.* Spatially-explicit footprints of agricultural commodities: Mapping carbon emissions embodied in Brazil’s soy exports. *Global Environmental Change* **62**, 102067 (2020).

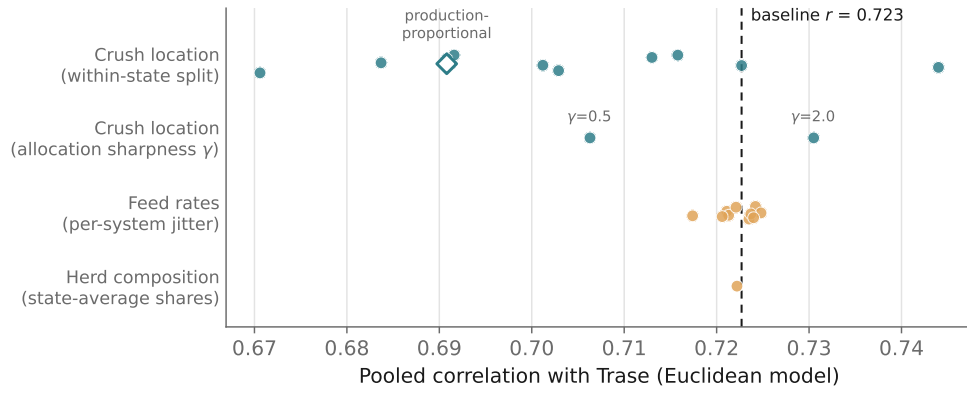


Fig. 1 Input-assumption sensitivity of the model–Trase correlation. Each dot is one perturbed re-run of the 2019 flow reconstruction, scored against Trase on the pooled municipality×destination correlation (Euclidean model); the dashed line marks the baseline run. Crush-location variants (teal) span the within-state capacity splits (ten seeded random draws plus the production-proportional split, open diamond) and the allocation-sharpness exponents $\gamma \in \{0.5, 2\}$; feed variants (orange) span ten seeded per-system feed-rate jitters and the state-average herd composition. Vertical scatter within a row is jitter for legibility only.